HUMAN ANATOMY & PHYSIOLOGY

THE EAR — Clear, Print-Ready Notes

Definition and functions

The ear is the organ of hearing and balance. It receives sound waves, converts them into nerve impulses, and helps the brain interpret sound. The inner ear also helps maintain balance and body position.

The ear is divided into three parts

• **1. External ear (outer ear):** Collects sound waves and directs them toward the eardrum.

• **2. Middle ear:** Contains three tiny bones that transmit and amplify vibrations.

• **3. Inner ear:** Contains the cochlea for hearing and the vestibular organs for balance.

1. External ear

The external ear consists mainly of the pinna (auricle) and the external auditory canal (external acoustic meatus).

Pinna (auricle)

The pinna is the visible part of the ear. It is made mainly of elastic cartilage covered by skin and helps collect sound waves and direct them into the auditory canal. The lower soft part is called the lobule or earlobe.

External auditory canal

The external auditory canal is about 2.5 cm long in adults. It carries sound waves from the pinna to the tympanic membrane (eardrum). Hair and ceruminous glands are present near the outer part of the canal.

Ceruminous glands and ear wax

Ceruminous glands produce cerumen (ear wax). Ear wax helps trap dust and small particles and protects the ear canal. It also contains substances that help protect against microbes.

Tympanic membrane (eardrum)

The tympanic membrane is a thin membrane at the end of the external auditory canal. Sound waves make it vibrate.

2. Middle ear

The middle ear is an air-filled cavity in the temporal bone, between the eardrum and the inner ear. It contains three small bones called auditory ossicles.

The three auditory ossicles

• **Malleus (hammer):** Its handle is attached to the tympanic membrane.

• **Incus (anvil):** Connects the malleus to the stapes.

• **Stapes (stirrup):** The smallest of the three bones; its footplate contacts the oval window.

These bones transmit vibrations from the eardrum to the oval window of the inner ear and help amplify sound.

Eustachian (auditory / pharyngotympanic) tube

The Eustachian tube connects the middle ear to the nasopharynx. It helps equalise air pressure on both sides of the tympanic membrane. It usually opens briefly during swallowing, yawning, or chewing.

3. Inner ear

The inner ear lies within the temporal bone and is also called the labyrinth. It contains the sensory organs for hearing and balance.

The labyrinth has two main parts

• **Bony labyrinth:** A system of cavities in bone, filled with perilymph around the membranous labyrinth.
• **Membranous labyrinth:** A system of membranous sacs and ducts filled with endolymph.

The inner ear has three main structures

1. **Cochlea** — hearing
2. **Vestibule** — senses head position and linear acceleration
3. **Semicircular canals** — sense rotational movement of the head

Cochlea

The cochlea is a spiral, snail-shell-shaped structure. It contains the sensory organ of hearing (the organ of Corti). Its fluid movements stimulate sensory hair cells, which convert vibrations into electrical nerve signals.

The cochlear duct divides the cochlea into three fluid-filled chambers:

• **Scala vestibuli:** filled with perilymph.
• **Scala media (cochlear duct):** filled with endolymph.
• **Scala tympani:** filled with perilymph.

Vestibule

The vestibule is the central part of the inner ear and connects with the cochlea and semicircular canals. It contains the utricle and saccule, which help detect head position and linear acceleration.

Semicircular canals

There are three semicircular canals: superior (anterior), posterior, and lateral (horizontal). They are arranged approximately at right angles to one another. Each has a widened region called the ampulla, containing sensory receptors that detect rotational movement and help maintain dynamic balance.

Physiology of hearing — pathway

1. The pinna collects sound waves and directs them into the external auditory canal.

2. Sound waves strike the tympanic membrane, causing it to vibrate.

3. The malleus, incus, and stapes transmit and amplify the vibrations.

4. The stapes moves at the oval window and creates pressure waves in the cochlear fluid.

5. Fluid movement causes movement of the basilar membrane and stimulates hair cells in the organ of Corti.

6. Hair cells convert mechanical movement into electrical nerve impulses.

7. The cochlear branch of the vestibulocochlear nerve (cranial nerve VIII) carries the signals to the brain, where sound is interpreted.

Balance

The vestibule and semicircular canals contain sensory receptors that send information about head position and movement through the vestibular branch of cranial nerve VIII. The brain uses this information, together with information from the eyes and muscles, to help maintain posture and balance.

Important terms for revision

Pinna — collects sound waves
Auditory canal — carries sound to the eardrum
Tympanic membrane — vibrates in response to sound
Ossicles — malleus, incus, stapes
Eustachian tube — equalises middle-ear pressure
Cochlea — hearing
Vestibule — head position / linear acceleration
Semicircular canals — rotational movement / dynamic balance
Cranial nerve VIII — vestibulocochlear nerve

Study tip: Learn the three parts of the ear first, then practise the pathway of hearing in order.

HUMAN ANATOMY & PHYSIOLOGY

THE EAR — Additional Notes (Clear, Print-Ready)

Transmission of Hearing Impulses

The vibrations in the cochlea stimulate sensory hair cells. These cells convert the mechanical movement into electrical nerve impulses. The impulses are carried by the cochlear branch of the vestibulocochlear nerve (cranial nerve VIII) to the brain.

Auditory Pathway

Cochlea (hair cells) → Cochlear branch of cranial nerve VIII → Brainstem pathways → Thalamus → Auditory area of the temporal lobe of the cerebral cortex.

Note: The handwritten page mentions the medulla and midbrain. The pathway above is presented in clear print-ready wording; the exact pathway involves several brainstem relay nuclei.

Next Topic

Sensation of smell (olfaction)

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